INTERNATIONAL DT466E SPECIFICATIONS



Engine Model: International Navistar DT466E

Engine Type: 4-cycle inline 6-cylinder turbocharged liquid-cooled diesel

Total Displacement: 466 cu.in (7.6 L)

Rated Engine Power: 175-275 Hp (130-205 kW) at 2300 rpm

Fuel System Type: Hydraulic electronic unit injector (HEUI)

- INTERNATIONAL NAVISTAR DT466E SPECIFICATIONS
- INTERNATIONAL NAVISTAR DT466E TROUBLESHOOTING

GENERAL TECHNICAL DATA

The International Navistar DT466E is an advanced evolution of the legendary DT466 engine series, engineered by Navistar as part of the electronically controlled medium-duty diesel family. Introduced in the mid-1990s, the DT466E incorporates the Hydraulically Actuated Electronically Controlled Unit Injector (HEUI) fuel system, giving it improved fuel efficiency, lower emissions, smoother idle, and more precise fuel delivery compared to its mechanically injected predecessors. The DT466E quickly became a workhorse for medium-duty trucks, school buses, municipal equipment, agricultural machinery, and industrial power units.

Like the earlier DT466 engines, the DT466E retains the core design philosophy of durability, serviceability, and rebuildability. This includes its wet-sleeve cylinder design, gear-driven valve train, robust cast-iron block, and excellent thermal efficiency. Fleet operators valued the DT466E for its long service intervals, ability to withstand harsh operating conditions, and cost-effective in-frame overhauls. The combination of mechanical robustness and electronic fuel control made the DT466E one of the most reliable engines of its class.

The Navistar International DT466E is a four-stroke, water-cooled, inline six-cylinder diesel engine. Total displacement is 7,638 cc (466 cubic inches). The bore measures 116 mm (4.59 inches), and the stroke is 119 mm (4.68 inches), resulting in a long-stroke configuration that favors high torque and efficient low-speed operation. Compression ratio varies by calibration but ranges between 16.0:1 and 17.5:1.

Power and torque outputs vary by application:
- DT466E (early electronic versions): 175-230 Hp (130-172 kW) at 2,200-2,600 rpm
- Turbocharged and intercooled versions: 215-260 Hp (160-194 kW)
- High-output vocational calibrations: up to 275 Hp (205 kW)

Peak torque for most versions falls between: 750 and 900 Nm (553-664 lb-ft) at approximately 1,300-1,500 rpm. The firing order is 1-5-3-6-2-4. Idle speed generally ranges from 650-750 rpm. Governed maximum speed is usually set between 2,600 and 2,800 rpm depending on vehicle application and emissions requirements.

CYLINDER BLOCK

The DT466E cylinder block is a heavy-duty cast-iron structure with the same replaceable wet-sleeve design used throughout the DT series. These sleeves are cooled on their outer surfaces by engine coolant, ensuring uniform temperature and reducing cylinder distortion.

The block is fully rebuildable, allowing in-frame overhauls using standardized parts and procedures. Thick main-bearing saddles and deep structural ribs reinforce the crankcase, enabling the engine to handle high cylinder pressures from turbocharged and electronically controlled combustion.

CYLINDER HEAD AND VALVES

ALL MANUFACTURERS

- Honda
- Kawasaki
- Kohler
- Kubota
- Mercury
- Yamaha
- BMW
- Toyota
- Mitsubishi
- Nissan
- Yanmar
- Cummins
- Deutz
- Caterpillar
- Isuzu
- Perkins
- Mazda
- Opel
- Saab
- Subaru
- Suzuki
- Volkswagen
- Audi
- Volvo
- Renault
- Chevrolet
- Volvo Penta
- John Deere
- International

The cylinder head is cast iron with a crossflow design for improved airflow. Depending on the generation, the engine may have two or four valves per cylinder; the majority of DT466E engines use four-valve heads for enhanced breathing, reduced emissions, and improved thermal efficiency. The valvetrain is actuated by a gear-driven camshaft located in the block, using pushrods and rocker arms to operate the valves.

Valve clearances (cold):
- Intake: 0.25-0.30 mm (0.010-0.012 in)
- Exhaust: 0.30-0.36 mm (0.012-0.014 in)

FUEL SYSTEM

The HEUI (Hydraulically Actuated Electronically Controlled Unit Injector) system is the hallmark feature of the DT466E. This system uses engine oil as the hydraulic medium to actuate the injectors, while electronic commands from the ECM control injection timing, duration, and quantity.

Key features include:
- High-pressure oil pump (HPOP) delivering regulated oil pressure
- Electronically controlled injectors
- Low-pressure fuel supply pump
- High-capacity fuel filtration

Injection pressures can exceed 1,200 bar depending on ECM calibration and oil pressure. The HEUI system greatly improves fuel atomization, emissions, and throttle response compared to mechanical pumps.

Fuel system maintenance:
- Replace fuel filters every 10,000-15,000 km or 250-300 operating hours
- Clean or replace injector O-rings as required
- Inspect HPOP and oil rail seals during major service
- Drain water separator as needed

LUBRICATION SYSTEM

The DT466E uses a full-pressure lubrication system with a gear-driven oil pump. The system not only lubricates engine components but also powers the HEUI injectors. This makes oil quality critical for proper engine performance.

Oil capacity: 18-22 liters (19-23 US quarts)
Normal oil pressure at operating speed: 300-450 kPa (44-65 psi)
Idle pressure: 100-150 kPa (15-22 psi)

Recommended oil viscosity includes SAE 15W-40 or 10W-30 diesel-rated oil meeting API CF, CG-4, CH-4, or CI-4 specifications depending on model year.

COOLING SYSTEM

The cooling system is a closed-loop design using a belt-driven centrifugal water pump. Coolant flows through the block, wet sleeves, and cylinder head, then through a radiator or remote-mounted cooling package depending on application.

Thermostat opening temperature: approximately 82°C (180°F)
Full-open temperature: 95°C (203°F)
Cooling capacity: 20-25 liters (21.1-26.4 US quarts)
Standard system pressure: 70-90 kPa (10-13 psi)

Intercooled turbocharged models include charge-air coolers to reduce intake-air temperature and increase combustion efficiency.

INTAKE AND EXHAUST SYSTEMS

Air intake varies by application:
- Trucks and buses: dry-element air filters
- Construction equipment: cyclone pre-cleaners or oil-bath systems

Turbocharged versions use an exhaust-driven turbocharger connected to a cast-iron manifold. Many DT466E engines incorporate an intercooler to increase air density and reduce combustion temperatures.

Exhaust systems may include:
- Mufflers
- Catalysts

Ford
Hyundai
Chrysler
Mercedes Benz
Evinrude
Johnson
Tohatsu
Detroit Diesel
Tecumseh
Briggs Stratton

- Backpressure regulation devices
- Electronic exhaust temperature monitoring

ELECTRICAL SYSTEM

The DT466E uses a 12- or 24-volt electrical system depending on vehicle specifications. Electronic engine management monitors and controls injection timing, fuel delivery, turbo boost (if equipped with VGT), oil pressure, coolant temperature, and numerous other parameters.

Alternator output: 80-140 amperes
Starter rating: 3.5-5.0 kW

Sensors include:
- Camshaft position
- Crankshaft position
- MAP/boost pressure
- Intake-air temperature
- Oil pressure and oil temperature
- Coolant temperature

OTHER SYSTEMS

Depending on installation, the DT466E can support:
- Air compressors for air brakes
- Hydraulic pump drives
- PTO drives
- Transmission-integrated control modules
- Remote cooling packages
- Generator heads

DIMENSIONS AND WEIGHT

Overall Length: 1,050-1,150 mm (41.3-45.3 in)
Overall Width: 720-820 mm (28.3-32.3 in)
Overall Height: 1,000-1,100 mm (39.4-43.3 in)
Dry weight: 1,000-1,100 kg (2,205-2,425 lb) depending on configuration

MAINTENANCE DATA

- Oil and filter: every 5,000-7,500 km or 200-250 hours
- Fuel filters: every 10,000-15,000 km or 250-300 hours
- Air filter: inspect every 50 hours
- Coolant: replace every 2 years
- Valve-lash adjustment: every 15,000-20,000 km
- Injector O-ring/HEUI inspection: every major service interval
- Compression test: every 100,000 km
- Drive-belt inspection: each service cycle

TIGHTENING TORQUES

- Cylinder head bolts: 210-240 Nm (155-177 lb-ft)
- Main bearing caps: 190-220 Nm (140-162 lb-ft)
- Connecting-rod bolts: 100-120 Nm (74-89 lb-ft)
- Flywheel bolts: 180-210 Nm (133-155 lb-ft)
- Crankshaft pulley bolt: 350-380 Nm (258-280 lb-ft)
- Oil drain plug: 40-55 Nm (30-41 lb-ft)
- Injector hold-downs: 25-30 Nm (18-22 lb-ft)

MACHINERY APPLICATIONS

The Navistar DT466E is widely used in:
- International/Navistar medium-duty trucks
- School buses
- Construction loaders and equipment
- Agricultural tractors and machinery
- Municipal, fleet, and utility vehicles
- Industrial generators and auxiliary power units

INDUSTRIAL ENGINES TECHNICAL SPECIFICATIONS

KUBOTA D750	KUBOTA D905	KUBOTA D1005
CAT 3056E	CAT 3064T	CAT 3116T
DEUTZ BF3L2011	DEUTZ BF4L1011F	DEUTZ BF4L2011
VOLVO PENTA D11-725	VOLVO PENTA D16-MH	VOLVO PENTA D2-75F
JOHN DEERE 4039D	JOHN DEERE 4039T	JOHN DEERE 4045D
INTERNATIONAL BD154	INTERNATIONAL D179	INTERNATIONAL D239
DETROIT DIESEL 12V71T	DETROIT DIESEL DD5	DETROIT DIESEL DD13
VOLVO PENTA D3-160	VOLVO PENTA D4-300	VOLVO PENTA D6-310
JOHN DEERE 3029D	JOHN DEERE 3029T	JOHN DEERE 4024T
INTERNATIONAL D282	INTERNATIONAL DT360	INTERNATIONAL DT466
DETROIT DIESEL 4-71	DETROIT DIESEL 6V53	DETROIT DIESEL 8V92TA
YANMAR 3TN75RJ	YANMAR 3TNA82	YANMAR 3TNE82A
CUMMINS 4BT3.9-G2	CUMMINS 4BTAA3.9	CUMMINS 6BTA5.9
ISUZU 4FC1	ISUZU 4HF1	ISUZU 4HJ1
MITSUBISHI D06FRC-TAA	MITSUBISHI K3E	MITSUBISHI L3E
CAT 3114	CAT 3116TA	CAT 3176B
INTERNATIONAL BD144	INTERNATIONAL D155	INTERNATIONAL D246
JOHN DEERE 4045DF150	JOHN DEERE 4045HF275	JOHN DEERE 4219DF
MITSUBISHI 4D34T	MITSUBISHI 6D22T	MITSUBISHI D04EG-T
CAT 3204	CAT 3208T	CAT 3306TA
INTERNATIONAL D268	INTERNATIONAL D358	INTERNATIONAL D360
MITSUBISHI S2E2	MITSUBISHI S3L2	MITSUBISHI S4E
JOHN DEERE 4239D	JOHN DEERE 4276T	JOHN DEERE 6059
CAT 3306B	CAT 3306 DITA	CAT 3408E

INTERNATIONAL D436	INTERNATIONAL DT407	INTERNATIONAL DT414
JOHN DEERE 6068T	JOHN DEERE 6068TF220	JOHN DEERE 6076A
MITSUBISHI S4S	MITSUBISHI S6E2	MITSUBISHI S6S
PERKINS 404C-22	PERKINS 404D-22T	PERKINS 1004-4T
CAT 3126B	CAT 3306B	CAT 3512B
DEUTZ BF4L913	DEUTZ BF6L913C	DEUTZ BF6M2012C
KUBOTA D1105T	KUBOTA D1402	KUBOTA D1503
YANMAR 3TNE88	YANMAR 3TNV74F	YANMAR 3TNV86CT
CUMMINS ISX15	CUMMINS QSB5.9	CUMMINS QSF2.8
PERKINS 1104C-44TA	PERKINS 1104D-E44TA	PERKINS 1106A-70TAG2
ISUZU 4BA1	ISUZU 4BD2	ISUZU 4BE2
KUBOTA V1305	KUBOTA V1702	KUBOTA V2607
YANMAR 4TN84T	YANMAR 4TNE92	YANMAR 4TNE94
DEUTZ D914L04	DEUTZ F4L2011	DEUTZ F4L912
CUMMINS ISC-285	CUMMINS ISF3.8	CUMMINS ISM-370
CAT C1.1	CAT C1.7	CAT C2.8
ISUZU 6BG1-T	ISUZU 6HE1-T	ISUZU 10PD1
PERKINS 1006-60	PERKINS 1103A-33G	PERKINS 1104A-44TG2
KUBOTA V3307-DI-T	KUBOTA Z500	KUBOTA Z1100
YANMAR 4TNV84T	YANMAR 4TNV94	YANMAR 4TNV98
CAT C3.4B	CAT C4.4 ACERT	CAT C15-6NZ
DEUTZ F6L914	DEUTZ F8L413F	DEUTZ TCD7.8L6
CUMMINS 6CTA8.3	CUMMINS 6LTAA8.9-G2	CUMMINS ISB4.5
ISUZU 4JG1-T	ISUZU 4LE1	ISUZU 6BF1
PERKINS 4.154	PERKINS T4.236	PERKINS 403D-07

YANMAR 2GM20F	YANMAR 2QM20	YANMAR 2TR13A
VOLVO PENTA D3-150	VOLVO PENTA D6-350	VOLVO PENTA D6-480
DEUTZ BF4M2011	DEUTZ BF6L914C	DEUTZ BF6M1012E
ISUZU 4EE1	ISUZU 4JJ1-TC	ISUZU 4JK1-TC
JOHN DEERE 3029DF120	JOHN DEERE 3029HFG03	JOHN DEERE 3164D
MITSUBISHI 4D32	MITSUBISHI 4DQ5	MITSUBISHI 4DR7
CAT 3066	CAT 3412C	CAT 3508
DETROIT DIESEL 3-53T	DETROIT DIESEL 4-53	DETROIT DIESEL 6-71T
CUMMINS L9-350	CUMMINS L10-330E	CUMMINS NTA855-G3
DEUTZ BF4M1011F	DEUTZ BF4M1012E	DEUTZ BF4M1013EC
ISUZU 3KC2	ISUZU 3KR1	ISUZU 3LD2
JOHN DEERE 4045DF270	JOHN DEERE 4045HF120	JOHN DEERE 4045HFC04
YANMAR 3GM30FC	YANMAR 3HM35F	YANMAR 3JH2-TE
MITSUBISHI 6D16	MITSUBISHI 6D22	MITSUBISHI 6D34
CAT 3516E	CAT 3512C	CAT C2.4
DETROIT DIESEL 6V53N	DETROIT DIESEL 6V92TA	DETROIT DIESEL 8V53
CUMMINS ISX400ST	CUMMINS ISX435	CUMMINS ISX550
DEUTZ BF3M1011F	DEUTZ BF4L914	DEUTZ BF4L1011FT
YANMAR 2TR20A	YANMAR 2V78C	YANMAR 2YM20
INTERNATIONAL D350	INTERNATIONAL DT402	INTERNATIONAL DT436
ISUZU 3AB1	ISUZU 3CD1	ISUZU 3CH1
JOHN DEERE 3179DF	JOHN DEERE 4024HF295	JOHN DEERE 4039DF001
VOLVO PENTA D7A-T	VOLVO PENTA D8-600	VOLVO PENTA MD17C
YANMAR 3JH3E	YANMAR 3JH5E	YANMAR 3JH30A

CAR ENGINES TECHNICAL SPECIFICATIONS

MITSUBISHI 3G83	MITSUBISHI 4A92	MITSUBISHI 4B11T
NISSAN A14	NISSAN CD20T	NISSAN E16S
TOYOTA 1AR-FE	TOYOTA 1AZ-FSE	TOYOTA 1CD-FTV
BMW M50B24TU	BMW N43B20	BMW M42B18
HONDA A20A4	HONDA B16A2	HONDA B18B4
MITSUBISHI 4D36	MITSUBISHI 4D68	MITSUBISHI 4G18
TOYOTA 1GD-FTV	TOYOTA 1NR-FE	TOYOTA 1TR-FE
NISSAN CA20E	NISSAN CR12DE	NISSAN HR16DE
HONDA D14Z6	HONDA D15B2	HONDA D15Z4
BMW M43B16	BMW M54B30	BMW M20B25
HONDA B18A1	HONDA B20A5	HONDA D13B2
NISSAN CA18DET	NISSAN CG10DE	NISSAN NA20
MAZDA B6T	MAZDA FS-DE	MAZDA KF-ZE
OPEL A17DTR	OPEL A18XER	OPEL A20NFT
SAAB B201	SAAB B202L	SAAB B204R
SUBARU EG33	SUBARU EJ20T	SUBARU EJ204
SUZUKI 1.6 DDIS	SUZUKI 1.9 DDIS	SUZUKI D16AA
VOLKSWAGEN AAZ	VOLKSWAGEN AEX	VOLKSWAGEN CAVD
AUDI ABC	AUDI ABK	AUDI ADU
VOLVO B230FB	VOLVO B234F	VOLVO B4164T
RENAULT B4DA	RENAULT H4BT	RENAULT H5DT
CHEVROLET LQ9	CHEVROLET LS3	CHEVROLET LT4
FORD 4CC	FORD 4EB	FORD M1DA
HYUNDAI D4DD	HYUNDAI G4CP	HYUNDAI G4DJ

CHRYSLER 420A	CHRYSLER A588	CHRYSLER EDZ
MERCEDES OM460LA	MERCEDES OM612	MERCEDES OM642
MAZDA L3-VDT	MAZDA L8-DE	MAZDA LF-DE
OPEL A10XEP	OPEL A13DTE	OPEL A16XER
SAAB B205E	SAAB B207R	SAAB B234I
SUBARU EA71	SUBARU EE20Z	SUBARU EF12
SUZUKI F5A	SUZUKI F9Q	SUZUKI F10D
AUDI AEB	AUDI BBJ	AUDI BCY
VOLVO B18FT	VOLVO B200E	VOLVO B204FT
RENAULT C2J	RENAULT D4F	RENAULT D7D
CHEVROLET LBZ	CHEVROLET LDE	CHEVROLET LM7
FORD P4JD	FORD SEBA	FORD T1DA
HYUNDAI D3EA	HYUNDAI D4BA	HYUNDAI D4CB
CHRYSLER EGG	CHRYSLER ERB	CHRYSLER ESF
MERCEDES BENZ M112	MERCEDES BENZ M156	MERCEDES BENZ M256
MITSUBISHI 4G32	MITSUBISHI 4G61	MITSUBISHI 4G63T
TOYOTA 1GZ-FE	TOYOTA 1JZ-FSE	TOYOTA 1KZ-TE
BMW M40B16	BMW M21D24	BMW M52B24TU
HONDA D16A9	HONDA F22A2	HONDA J30AC
TOYOTA 2AD-FHV	TOYOTA 2GR-FSE	TOYOTA 3S-GE
MITSUBISHI 4G92A	MITSUBISHI 4J10	MITSUBISHI 4N14
NISSAN LD23	NISSAN MR18DE	NISSAN MR20DD
VOLVO B4184S2	VOLVO B4194T	VOLVO B4204T23
MERCEDES M113K	MERCEDES M266	MERCEDES OM647
NISSAN L18	NISSAN L20B	NISSAN L28ET

VOLVO B20E	VOLVO B200ET	VOLVO B230K

SMALL ENGINES TECHNICAL SPECIFICATIONS

KAWASAKI FC290V	KAWASAKI FD501V	KAWASAKI FD620D
HONDA G65	HONDA G80	HONDA G150
KOHLER CH680	KOHLER CH750S	KOHLER CV13S
TECUMSEH HS50	TECUMSEH HSSK50	TECUMSEH OH160
BRIGGS 282707	BRIGGS 311707	BRIGGS 31G777
KAWASAKI FD661D	KAWASAKI FD711V	KAWASAKI FH680V
HONDA G300	HONDA GC135	HONDA GCV170
KOHLER CV15S	KOHLER CV745	KOHLER CV491S
KAWASAKI FH601V	KAWASAKI FH531V	KAWASAKI FH721V
HONDA GV200	HONDA GX31	HONDA GX120
KOHLER ECV749	KOHLER ECV870	KOHLER ECV940
KAWASAKI FR651V	KAWASAKI FR730V	KAWASAKI FS730V
HONDA GCV520	HONDA GP160	HONDA GSV190
KOHLER CH15S	KOHLER CH22S	KOHLER CH640S
KAWASAKI FX481V	KAWASAKI FX600V	KAWASAKI FX801V
TECUMSEH H70	TECUMSEH HH60	TECUMSEH HM80
BRIGGS 31H777	BRIGGS 31P977	BRIGGS 31Q777
HONDA GX240	HONDA GX360	HONDA GXV120
KOHLER ECH440	KOHLER ECH650	KOHLER ECH749
KAWASAKI FX1000V	KAWASAKI FS691V	KAWASAKI FX921V
KOHLER K241	KOHLER SV601	KOHLER SV610
HONDA GXV140	HONDA GXV270	HONDA GXV630

OUTBOARD ENGINES TECHNICAL SPECIFICATIONS

YAMAHA 9.9FMH	YAMAHA F4BMH	YAMAHA 15FMHS
MERCURY 150 XR6	MERCURY 225 SALTWATER	MARINER 150 MAGNUM
YAMAHA F5AMH	YAMAHA 9.9MSH	YAMAHA F6CMH
MERCURY 20 JET	MARINER 25 MARATHON	MERCURY 25 SEAPRO
EVINRUDE FICHT 115	EVINRUDE INTRUDER 150	EVINRUDE OCEAN PRO150
JOHNSON RUNNER 200	JOHNSON SEAHORSE 35	JOHNSON TRACKER 40
TOHATSU M25C2	TOHATSU M30A3	TOHATSU MD40B2
YAMAHA 20DMH	YAMAHA 30DETO	YAMAHA 25MLH
MERCURY 150 EFI	MARINER 175 MAGNUM	MERCURY 175 XRI
YAMAHA 30DMO	YAMAHA 25DM	YAMAHA 30AE

Gasoline and Diesel Engines Technical Data